

Section C

Answer any two questions.

Each question carries 10 marks.

24. (a) State six standard methods for proving theorems and briefly explain any two of them with the help of examples.
- (b) Using the laws of logic simplify the Boolean Expression $(p \wedge \neg q) \vee q \vee (\neg p \wedge q)$.
25. (a) Prove that there is no polynomial $f(n)$ with integral coefficients that will produce primes for all integers n .
- (b) State the prime number theorem and find six consecutive integers that are composites.
26. (a) State and prove Fundamental Theorem of Arithmetic.
- (b) Find the largest power of 3 that divides 207!
27. (a) Let p be a prime and a any integer such that $p \nmid a$. Then show that the least residues of the integers $a, 2a, 3a, \dots, (p-1)a$ modulo p are a permutation of the integers $1, 2, 3, \dots, (p-1)$.
- (b) Find the remainder when 24^{1947} is divided by 17.

(2 × 10 = 20 marks)

**FIRST SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

ME 1C 01—MATHEMATICAL ECONOMICS

(2016—2018 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

ME 1C 01—MATHEMATICAL ECONOMICS
(Multiple Choice Questions for SDE Candidates)

1. The demand for essential good is :
(A) Elastic. (B) Inelastic.
(C) Relatively elastic. (D) Relatively inelastic.
2. The ratio of the percentage change in sales of one product to the percentage change in price of another product is called :
(A) Arc price elasticity. (B) Point price elasticity.
(C) Cross price elasticity. (D) Income elasticity.
3. Perfectly elastic supply curve is :
(A) Parallel to X axis.
(B) Parallel to Y axis.
(C) Sloping curve passing through the origin.
(D) None of the these.
4. The variables in the demand function which are related to price are :
(A) Own price of the product. (B) Price of compliment.
(C) Price of substitutes. (D) All the above.
5. When the purchases of goods increase with rising levels of income, such goods are called :
(A) Inferior goods. (B) Normal goods.
(C) Giffen goods. (D) Laxurious goods.
6. In a demand curve price is measured along the :
(A) Vertical axis. (B) Horizontal axis.
(C) Both (A) and (B). (D) None.
7. A shift of the whole demand curve to the left indicates :
(A) An increase in demand. (B) Decrease in demand.
(C) Demand is constant. (D) None.

8. The average responsiveness of the dependent variable to changes in the independent variable over some interval is measured by :
- (A) Point elasticity. (B) Arc elasticity.
(C) Supply elasticity. (D) None.
9. When the cross price elasticity $e_c = 0$, the goods are ?
- (A) Substitutes. (B) Complements.
(C) Normal. (D) Independent.
10. For a less elastic supply curve, η_p^s is :
- (A) Less than 1. (B) More than 1.
(C) Equal to 1. (D) Zero.
11. The total of the quantities demanded by all consumers in an economy at each price is called :
- (A) Market demand curve. (B) Market supply curve.
(C) Market equilibrium. (D) None of these.
12. When $|e_p| < 1$, the demand is :
- (A) Elastic. (B) Inelastic.
(C) Unitarily elastic. (D) None.
13. A given percentage change in price results in an equal percentage change in sales, indicates :
- (A) Unitary price elasticity. (B) Inelastic price elasticity.
(C) Elastic price elasticity. (D) None.
14. Profit is equal to total revenue minus :
- (A) Explicit costs. (B) Implicit costs.
(C) Implicit costs and explicit costs. (D) Wages and rents.
15. The elasticity of demand η_d in terms of AR and MR is :
- (A) $\frac{AR - MR}{AR}$. (B) $\frac{AR - MR}{MR}$.
(C) $\frac{MR}{AR - MR}$. (D) $\frac{AR}{AR - MR}$.

16. A distinction between cost of production and expenses of production is made by :
- (A) Engel. (B) Marshall.
(C) Keynes. (D) None of these.
17. When marginal cost is greater than average cost, the total cost elasticity will be :
- (A) Greater than 1. (B) Less than 1.
(C) Equal to 1. (D) None.
18. The equation $\eta_p = \frac{AR}{AR - MR}$ indicates that marginal revenue as a function of :
- (A) Elasticity of demand. (B) Average revenue.
(C) Both (A) and (B) and None. (D) None.
19. The first order condition for utility maximization gives :
- (A) $\frac{MU_1}{MU_2} = \frac{P_1}{P_2}$. (B) $\frac{MU_1}{P_2} = \frac{MU_1}{P_1}$.
(C) $\frac{MU_2}{MU_1} = \frac{P_1}{P_2}$. (D) None of the above.
20. The point at which the marginal utility first increases, reaches the maximum, then diminishes is called :
- (A) Point of inflexion. (B) Minimum point.
(C) Saturation point. (D) None of these.

**FIRST SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

ME 1C 01—MATHEMATICAL ECONOMICS

(2016—2018 Admissions)

Time : Three Hours

Maximum : 80 Marks

Part A

Answer all the twelve questions.

Each question carries 1 mark.

1. The demand for essential good is :
 - (a) Elastic.
 - (b) Inelastic.
 - (c) Relatively elastic.
 - (d) Relatively inelastic.
2. When the demand curve shifts to right, there occurs :
 - (a) An increase in demand.
 - (b) Decrease in demand.
 - (c) Demand is constant.
 - (d) None.
3. When the cross price elasticity is positive, the products are ?
 - (a) Substitutes.
 - (b) Compliments.
 - (c) Normal.
 - (d) None of these.
4. Profit is equal to total revenue minus :
 - (a) Explicit costs.
 - (b) Implicit costs.
 - (c) Implicit costs and explicit costs.
 - (d) Wages and rents.
5. Total variable cost plus total fixed cost gives :
 - (a) Total cost.
 - (b) Average cost.
 - (c) Marginal cost.
 - (d) None of these.

Turn over

6. The ratio of the proportionate change in average cost to the proportionate change in output is called elasticity of :
- (a) Marginal cost. (b) Total cost.
(c) Average cost. (d) None.
7. The rate at which the consumer trades off one commodity for another is called :
- (a) Marginal rate of technical substitution.
(b) Marginal rate of substitution.
(c) Equi-marginal utility.
(d) None of these.
8. The indifference curves for perfect substitutes are :
- (a) Straight lines. (b) L-shaped.
(c) Curves. (d) Concave from above.
9. An attribute possessed by a commodity to satisfy a human want, to yield satisfaction to consumer is termed as :
- (a) Utility. (b) Preference.
(c) Want. (d) None of these.
10. Let $50 + 10kk^2$ be a production function, where k represents capital. Then the marginal productivity when $k = 1$ is :
- (a) 116. (b) 16.
(c) 58. (d) 8.
11. For the function $y = 4x_1x_2 + x_1^3 + 2x_2^2$, the partial derivative $\frac{\partial y}{\partial x_1}$ is :
- (a) None of the following. (b) $4x_1 + 3x_1^2$.
(c) $4x_1 + 4x_2$. (d) $4x_2 + 3x_1^2$.
12. Behaviour of the function defined by $y = x^3 - 7x^2 + 6x - 2$ at $x = 4$ is :
- (a) Increasing. (b) Stationary.
(c) Decreasing. (d) None.

Part B

Answer any six questions in two or three sentences.

Each question carries 3 marks.

13. What is 'Demand Curve' ?
14. Explain briefly the 'Law of Supply'
15. What is 'Elasticity' ? Write any two types of elasticity.
16. What is the difference between cost of production and expenses of production ?
17. What is the nature of short term cost functions ?
18. Write a short note on marginal rate of substitution.
19. What are the maxima and minima conditions of consumer's equilibrium.
20. Find all the two first order partial derivatives of $z = 3x^2y^3$.
21. Find the marginal revenue function, given the average revenue function $AR = 10 - 0.5q$.

(6 × 3 = 18 marks)

Part C

Answer any six questions from the following.

Each question carries 5 marks.

22. What are the properties of price elasticity of demand ?
23. Give the nature and property of a demand function for a normal good.
24. Distinguish between point elasticity and arc elasticity.
25. Suppose the price p and the quantity q of a commodity is related by the equation $q = 30 - 4p - q^2$. Find the elasticity of demand at $p = 2$.
26. Explain the concept of rate of commodity substitution.
27. Find the maximum profit : Given $TR = 1400Q - 6Q^2$, $TC = 1500 + 80Q$.
28. Given $z = 8x^2 + 3y^2$, $x = 4t$, $y = 5t$. Find $\frac{dz}{dt}$.
30. Find the critical points, given $z = 2y^3 - x^3 + 147x - 54y + 12$.

(6 × 5 = 30 marks)

Turn over